FINITE OPTIMALITY AND DISCRETENESS FOR MULTIVARIABLE FRAME ENERGIES: NASH SELECTION AND THE SINGLE-SPHERE COMPARISON

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Abstract. We study two-point energies on products of real unit spheres, allowing correlated coordinates within the underlying probability measure. First, for every vector of positive rational exponents, each minimizer of the tensor-product p-frame energy admits a minimizing replacement supported on at most an explicit polynomial-feature dimension. This does not assert finite support of the original minimizer. We explain in detail why the rational finite-replacement proof uses no Nash curve selection, and why the Nash–Wronskian argument used for the stronger single-sphere denominator theorem has no immediate coordinatewise analogue in products. Second, for positive mixtures of powers we prove an exact quadratic-certificate theorem: a mean-exponent condition yields the minimum energy and, under a strict contact criterion, classifies every minimizer as a uniform measure on an orthonormal basis of product tensor directions, up to projective sign splitting. A sharp converse within factorized exponent mixtures constructs infinitely supported minimizers whenever a coordinate factor is purely quadratic. These conclusions include nontrivial rational and irrational superquadratic interactions but do not settle the classical pure $p > 2$ discreteness conjecture.

Document status. Preprint-style mathematical working draft; not submitted to arXiv, not peer reviewed, and not a claim of research priority. All assertions proved here are established without invoking results from the bibliography. The related single-sphere rational finite-replacement and denominator results belong to Park [8]; the latter is stated only as literature context, not used in any proof.

1. Introduction and main statements

Fix integers $m \geq 1$, $d_j \geq 2$, and put

$$X = \prod_{j=1}^{m} \mathbb{S}^{d_j-1}, \qquad H = \bigotimes_{j=1}^{m} \mathbb{R}^{d_j}, \qquad D = \dim H = \prod_{j=1}^{m} d_j. \tag{1.1}$$

Here $\mathcal{P}(X)$ denotes the Borel probability measures on X. The coordinates x_j may be arbitrarily dependent under $\mu \in \mathcal{P}(X)$; no product assumption on μ is made. Given $\mathbf{p} = (p_1, \dots, p_m) \in (0,\infty)^m$, we consider

$$K_{\mathbf{p}}(x,y) = \prod_{j=1}^{m} |\langle x_j, y_j\rangle|^{p_j}, \qquad \mathcal{E}_{\mathbf{p}}(\mu) = \iint_{X^2} K_{\mathbf{p}}(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y). \tag{1.2}$$

Our "multivariable" kernel acts on pairs of points in a *product of spheres*; it is distinct from the k-point, $k \geq 3$, multivariate potentials considered in [4, 5]. It is a natural tensor counterpart to the sphere energies studied in [1, 2, 3, 8].

For the single-sphere kernel $|\langle x, y\rangle|^p$, the conjecture of Bilyk, Glazyrin, Matzke, Park and Vlasiuk states that every minimizer is finitely supported if p is not an even positive integer [3]. Park's 2026 manuscript [8] establishes a finite replacement for all positive rational $p = a/b$, and finiteness of *every* minimizing measure when $b \geq d$. Our goal here is to separate three assertions: finite existence, finite replacement inside the original support, and finiteness of each original minimizing measure.

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Theorem 1.1 (Rational tensor finite replacement; proved in Section 3)**.** *Suppose $p_j = a_j/b_j$ are positive rational numbers in lowest terms. Set*

$$L = \operatorname{lcm}_j \begin{cases} b_j, & a_j \text{ even}, \\ 2b_j, & a_j \text{ odd}, \end{cases} \qquad n_j = Lp_j \in 2\mathbb{N}, \qquad R = \prod_{j=1}^{m} \binom{d_j + n_j - 1}{n_j}. \tag{1.3}$$

Every minimizing measure μ for (1.2) has a minimizing finitely supported replacement ν with $\operatorname{supp}\nu \subseteq \operatorname{supp}\mu$ and $\#\operatorname{supp}\nu \le R$. The two potentials agree on $\operatorname{supp}\mu$.

Theorem 1.1 includes rational exponents larger than 2, but its conclusion *does not say* that μ itself is finite. A second approach proves that stronger conclusion for a different and explicit family. For a finite positive measure σ on $(0,\infty)$, write

$$h_\sigma(t) = \int_{(0,\infty)} t^p \,\mathrm{d}\sigma(p), \quad 0 \le t \le 1, \qquad c_\sigma = \sigma((0,\infty)), \qquad \bar{p}_\sigma = c_\sigma^{-1} \int p \,\mathrm{d}\sigma(p). \tag{1.4}$$

The value t^p is defined to be 0 at $t = 0$ for every $p > 0$.

Theorem 1.2 (Sharp factorized-mixture classification; proved in Section 5)**.** *For each j let σ_j be a nonzero finite positive measure on $(0,\infty)$ with finite first moment and $\bar{p}_{\sigma_j} \le 2$. Put $c_j = c_{\sigma_j}$ and*

$$K(x,y) = \prod_{j=1}^{m} h_{\sigma_j}(|\langle x_j, y_j\rangle|). \tag{1.5}$$

Then

$$\min_{\mu\in\mathcal{P}(X)} \iint K(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) = \frac{\prod_{j=1}^{m} c_j}{D}. \tag{1.6}$$

Within this class, every *minimizer is finitely supported if and only if no σ_j is concentrated at $p = 2$. When this condition holds, every minimizing measure puts mass $1/D$ on each of precisely D mutually orthogonal projective product-tensor lines, and its support has size at most $2^m D$. If some σ_j is concentrated at 2, an infinitely supported minimizer exists.*

The mean-exponent condition is sharp for the pointwise quadratic certificate, and not asserted to characterize the minimum outside its stated hypotheses. Section 6 also treats *correlated* joint laws of exponent vectors. The elementary lemmas needed in all subsequent arguments are proved in Section 2.

2. Elementary foundations used throughout

We collect proofs of the finite-dimensional and measure-theoretic facts needed later. All ambient spaces are compact metric spaces. A Borel probability measure assigns total mass one; its *support* is the set of points every open neighborhood of which has positive mass. A sequence μ_n converges *weakly* to μ when $\int f\,\mathrm{d}\mu_n \to \int f\,\mathrm{d}\mu$ for every continuous real-valued f. A kernel K is *positive semidefinite on S* if every finite matrix $[K(x_i,x_k)]$ with $x_i \in S$ is positive semidefinite. For a finite signed measure η we abbreviate

$$I_K(\eta,\eta) = \iint K(x,y)\,\mathrm{d}\eta(x)\mathrm{d}\eta(y), \qquad I_K(\eta,\zeta) = \iint K(x,y)\,\mathrm{d}\eta(x)\mathrm{d}\zeta(y).$$

The latter are well defined for bounded continuous K.

Lemma 2.1 (Compactness and continuity of the energy)**.** *For a compact metric space Z, every sequence in $\mathcal{P}(Z)$ has a weakly convergent subsequence. If K is continuous on $Z \times Z$, the map $\mu \mapsto I_K(\mu,\mu)$ is weakly continuous. Consequently the energy attains its minimum over $\mathcal{P}(Z)$.*

Proof. We give an explicit construction of the limit measure. For each n choose a finite Borel partition $\mathcal{Q}_n$ of Z into sets of diameter at most $1/n$ (use a finite cover by sufficiently small open balls, taking successive set differences). Let $\mathcal{A}_n$ be the common refinement of $\mathcal{Q}_1,\dots,\mathcal{Q}_n$. These partitions are nested, have finitely many nonempty cells, and their cell diameters tend to zero. Given a sequence (μ_k), diagonal extraction gives a subsequence, again indexed by k,

such that $\mu_k(A) \to a(A)$ for every cell of every $\mathcal{A}_n$. The limits are nonnegative, sum to one at each level, and a parent-cell weight equals the sum of its child-cell weights.

Here is a direct construction of a probability measure with these limiting integrals, without an abstract representation theorem. Partition the interval $[0,1]$ into intervals of lengths $a(A)$ for cells A of $\mathcal{A}_1$; within each interval for a parent cell, subdivide into intervals of lengths $a(B)$ for its children at level 2, and continue. For Lebesgue-almost every $u \in [0,1]$, avoiding the countably many interval endpoints, this selects a nested sequence of cells $A_n(u)$ with diameters tending to zero. Choose a representative $z_A \in A$ for every nonempty cell and put $z_n(u) = z_{A_n(u)}$. Since $A_{n+1}(u) \subset A_n(u)$, the sequence $z_n(u)$ is Cauchy and converges to a point $z(u) \in Z$. It is measurable as the pointwise limit of finite-valued measurable functions. Let μ be the distribution of $z(u)$ when u is uniform on $[0,1]$.

For a continuous f, let $\omega_f(\delta)$ be its modulus of continuity. For every n, both $\int f \,\mathrm{d}\mu_k$ and $\int_0^1 f(z(u))\,\mathrm{d}u$ differ by at most $\omega_f(1/n)$ from the corresponding finite sums $\sum_{A\in\mathcal{A}_n} \mu_k(A) f(z_A)$ and $\sum_{A\in\mathcal{A}_n} a(A) f(z_A)$, respectively. The finite sums converge as $k \to \infty$, then $n \to \infty$ gives $\int f \,\mathrm{d}\mu_k \to \int f \,\mathrm{d}\mu$. This proves weak sequential compactness.

To verify energy continuity without assuming that product measures converge, fix $\delta > 0$ and select a finite δ-net $a_1, \dots, a_N$ in Z. Set $w_i(x) = \max(0, 2\delta - \operatorname{dist}(x, a_i))/\sum_h \max(0, 2\delta - \operatorname{dist}(x, a_h))$. The denominator is positive, each w_i is continuous, and $w_i(x) > 0$ implies $\operatorname{dist}(x, a_i) < 2\delta$. Therefore

$$K_\delta(x,y) = \sum_{i,h=1}^{N} K(a_i, a_h) w_i(x) w_h(y)$$

converges uniformly to $K(x,y)$ as $\delta \downarrow 0$, by uniform continuity of K. This is a finite sum of separated continuous functions. For such a sum,

$$I_{\sum_i f_i \otimes g_i}(\mu,\mu) = \sum_i \Big(\int f_i \,\mathrm{d}\mu\Big)\Big(\int g_i \,\mathrm{d}\mu\Big),$$

which is weakly continuous. The uniform approximation transfers the claim to K. Applying compactness to a minimizing sequence completes the proof. □

Lemma 2.2 (A continuous finite-dimensional Gram representation)**.** *Let S be compact and $K : S \times S \to \mathbb{R}$ continuous, symmetric and positive semidefinite. Suppose the ranks of all finite Gram matrices are bounded by R. Then there is $r \le R$ and a continuous map $\Phi : S \to \mathbb{R}^r$ such that $K(x,y) = \langle \Phi(x), \Phi(y) \rangle$ for all $x, y \in S$.*

Proof. Let r be the maximum rank of all finite Gram matrices; the maximum exists because these ranks are integers in $\{0, \dots, R\}$. Select r anchor points $a_1, \dots, a_r$ with invertible positive definite Gram matrix $G = [K(a_i, a_j)]$. Such anchors exist by choosing a maximal linearly independent subset of Gram vectors from any sample realizing rank r. Put

$$k(x) = (K(a_1, x), \dots, K(a_r, x))^{\mathsf{T}}, \qquad \Phi(x) = G^{-1/2} k(x).$$

The enlarged Gram matrix for the anchors together with any x, y still has rank r. Its Schur complement of G is therefore zero, so

$$K(x,y) = k(x)^{\mathsf{T}} G^{-1} k(y) = \langle \Phi(x), \Phi(y) \rangle.$$

Continuity follows from the displayed formula. For $r = 0$ the zero map into the zero-dimensional space suffices. □

Lemma 2.3 (Affine convex compression, with proof)**.** *Let $C \subset \mathbb{R}^r$ be compact and contained in an affine subspace of dimension at most q. If $\lambda \in \mathcal{P}(C)$, then its barycenter $v = \int_C u \,\mathrm{d}\lambda(u)$ is a convex combination of at most $q+1$ points of C.*

Proof. First, consider a finite convex combination with $N > q+1$ positive coefficients. Its N points are affinely dependent, so there exist real a_i, not all zero, with $\sum_i a_i = 0$ and $\sum_i a_i u_i = 0$. Since the a_i have both signs, subtracting $t a_i$ from each coefficient for the largest permissible $t > 0$ preserves nonnegativity, total mass and the barycenter, and makes at least

one coefficient zero. Repetition reduces every finite convex combination to at most $q+1$ points.

The set of these short combinations is compact: it is the continuous image of C^{q+1} times the compact probability simplex. Approximating the integral defining v by finite convex combinations (uniformly approximating the identity map on a finite partition of the compact set C) therefore shows that v itself is one of these short convex combinations. This is the claimed bound. □

Lemma 2.4 (Vanishing contact gaps on the support)**.** *Let $G : X \times X \to [0,\infty)$ be continuous and $\mu \in \mathcal{P}(X)$. If $\iint G \,\mathrm{d}\mu\mathrm{d}\mu = 0$, then $G(x,y) = 0$ for all $x, y \in \operatorname{supp}\mu$.*

Proof. If $G(x_0,y_0) > 0$ for support points x_0, y_0, continuity gives neighborhoods U, V and $\varepsilon > 0$ with $G \geq \varepsilon$ on $U \times V$. Then $\int G \,\mathrm{d}\mu\mathrm{d}\mu \geq \varepsilon\mu(U)\mu(V) > 0$, a contradiction. □

Lemma 2.5 (Strict exponential averaging)**.** *If Z is an integrable real random variable and $\mathbb{E}e^Z$ is defined in $[0,\infty]$, then $\mathbb{E}e^Z \geq e^{\mathbb{E}Z}$, with equality only if Z is constant almost surely.*

Proof. For every real w, $e^w \geq 1 + w$, with equality precisely when $w = 0$. Apply this to $w = Z - \mathbb{E}Z$, take expectations, and multiply by $e^{\mathbb{E}Z}$. If $\mathbb{E}e^Z$ is infinite the strict inequality is automatic. □

Lemma 2.6 (Product tensors, orthogonality, and projective signs)**.** *For unit vectors $x_j, y_j \in \mathbb{R}^{d_j}$ and $z(x) = x_1 \otimes \cdots \otimes x_m$, one has $\|z(x)\| = 1$ and*

$$\langle z(x), z(y)\rangle = \prod_{j=1}^{m} \langle x_j, y_j\rangle.$$

Moreover, $\mathbb{R}z(x) = \mathbb{R}z(y)$ if and only if $x_j = \pm y_j$ for every j. Thus a projective product-tensor line has at most 2^m representatives in X.

Proof. The tensor inner-product identity follows from its defining property on pure tensors. If the two unit tensors span the same line, their absolute inner product is 1; hence $\prod_j |\langle x_j, y_j\rangle| = 1$. Each factor is at most 1, so all factors equal 1, and equality in Euclidean Cauchy–Schwarz yields $x_j = \pm y_j$. The converse is immediate. □

3. Variational support positivity and rational finite rank

The following proof makes explicit why a kernel that is not positive semidefinite on the whole space becomes positive semidefinite when restricted to a minimizing support. Its single-sphere antecedent is in [8]; we include all steps to make the tensor argument self-contained.

Lemma 3.1 (First and second variations)**.** *Let $K : X \times X \to [0,\infty)$ be continuous and symmetric, with $K(x,x) = 1$ for all x. A minimizing $\mu \in \mathcal{P}(X)$ exists and its minimum e is positive. For $S = \operatorname{supp}\mu$ and $F_\mu(x) = \int K(x,y)\,\mathrm{d}\mu(y)$,*

$$F_\mu(x) \geq e \quad (x \in X), \qquad F_\mu(x) = e \quad (x \in S). \tag{3.1}$$

Moreover, for any points $x_1, \ldots, x_N \in S$ and coefficients $a_1, \ldots, a_N \in \mathbb{R}$,

$$\sum_{i,k=1}^{N} a_i a_k K(x_i, x_k) \geq e\Big(\sum_{i=1}^{N} a_i\Big)^2. \tag{3.2}$$

In particular, $K|_{S\times S}$ is positive semidefinite.

Proof. Lemma 2.1 proves attainment. Given any μ, take $x \in \operatorname{supp}\mu$. There is a neighborhood U of x with $K(u,v) > 1/2$ for $u, v \in U$, hence the energy is at least $\mu(U)^2/2 > 0$. Therefore the attained minimum e is positive.

Comparing μ with $(1-t)\mu + t\delta_x$ for $t \downarrow 0$ gives $F_\mu(x) \geq e$. Since $\int F_\mu \,\mathrm{d}\mu = e$, continuity forces equality on all of S.

For each x_i, normalize the restriction of μ to a shrinking neighborhood of x_i to obtain a measure $\mu_{i,\varepsilon} \Rightarrow \delta_{x_i}$. Put $A = \sum_i a_i$ and $\eta_\varepsilon = \sum_i a_i \mu_{i,\varepsilon} - A\mu$. This signed measure has zero

mass and bounded density with respect to μ. Thus $\mu + t\eta_\varepsilon$ is a probability measure for both signs of sufficiently small t. The linear variation vanishes because $F_\mu = e$ on S; minimality then gives $\iint K \,\mathrm{d}\eta_\varepsilon \mathrm{d}\eta_\varepsilon \geq 0$. Passing to the weak limit yields

$$0 \leq \iint K \,\mathrm{d}\Big(\sum_i a_i \delta_{x_i} - A\mu\Big)^2 = \sum_{i,k} a_i a_k K(x_i, x_k) - eA^2,$$

which proves (3.2). □

Lemma 3.2 (Rank monotonicity under integral Hadamard powers)**.** *If G is a real positive semidefinite matrix and $L \geq 1$ is an integer, then*

$$\operatorname{rank} G \leq \operatorname{rank}(G^{\circ L}),$$

where $G^{\circ L}$ is the entrywise Lth power.

Proof. Write $G_{ik} = \langle v_i, v_k \rangle$. Select $r = \operatorname{rank} G$ linearly independent Gram vectors $v_{i_1}, \ldots, v_{i_r}$. Choose linear functionals ℓ_s dual to these vectors on their span. Since $\ell_s^{\otimes L}(v_{i_t}^{\otimes L}) = \delta_{st}$, their tensor powers are also linearly independent. The latter are Gram vectors for $G^{\circ L}$, whose rank is consequently at least r. □

Proof of Theorem 1.1. Choose a minimizer μ, with support S, and let $K = K_{\mathbf{p}}$. By Lemma 3.1, every finite Gram matrix $G = [K(x_i, x_k)]$ sampled from S is positive semidefinite. Choose L as in (1.3). Since $n_j = Lp_j$ is even, the powered matrix $G^{\circ L}$ is the Gram matrix of

$$\Psi(x) = \bigotimes_{j=1}^m x_j^{\otimes n_j} \in \bigotimes_{j=1}^m \operatorname{Sym}^{n_j}(\mathbb{R}^{d_j}),$$

a vector space of dimension R. Indeed, $\operatorname{Sym}^n(\mathbb{R}^d)$ has a basis of symmetrized coordinate monomials $e_1^{\odot k_1} \odot \cdots \odot e_d^{\odot k_d}$ with nonnegative integers k_i summing to n; these monomials are linearly independent as homogeneous-polynomial coefficients. There are $\binom{d+n-1}{n}$ such multi-indices, by inserting $d-1$ separators among n identical symbols. Lemma 3.2 therefore implies $\operatorname{rank} G \leq R$ for *every* finite sample. Thus Lemma 2.2 supplies a continuous feature map $\Phi : S \to \mathbb{R}^r$, $r \leq R$, satisfying $K(x, y) = \langle \Phi(x), \Phi(y) \rangle$ on $S \times S$.

Let $M = \int_S \Phi(x) \,\mathrm{d}\mu(x)$. By (3.1),

$$\|M\|^2 = e > 0, \qquad \langle \Phi(x), M \rangle = e \quad (x \in S). \tag{3.3}$$

Hence $\Phi(S)$ lies in an affine hyperplane of dimension $r-1$. As M is the barycenter of a probability measure supported on this compact affine hyperplane, Lemma 2.3 yields $x_1, \ldots, x_s \in S$, $s \leq r \leq R$, and weights $w_i \geq 0$ summing to one with $\sum_i w_i \Phi(x_i) = M$. For $\nu = \sum_i w_i \delta_{x_i}$, equation (3.3) yields

$$\mathcal{E}_{\mathbf{p}}(\nu) = \Big\| \sum_i w_i \Phi(x_i) \Big\|^2 = e, \qquad F_\nu(x) = \langle \Phi(x), M \rangle = F_\mu(x) \quad (x \in S).$$

Thus ν is a minimizing replacement with $s \leq R$ atoms. □

Example 3.3 (Two rational exponents above 2)**.** For $d_1 = d_2 = 2$ and $(p_1, p_2) = (5/2, 3)$, one can take $L = 4$, $(n_1, n_2) = (10, 12)$, and

$$R = \binom{11}{10}\binom{13}{12} = 143.$$

Every minimizing measure has a minimizing replacement on at most 143 points within its support. The original minimizing measure is not asserted to be finite.

4. Tensor quadratic lower bounds and the contact structure

Define the rank-one unit tensor $z(x) = x_1 \otimes \cdots \otimes x_m \in H$. Its tensor second moment is the positive operator

$$T_\mu = \int_X z(x)z(x)^{\mathsf{T}} \, \mathrm{d}\mu(x), \qquad \operatorname{tr} T_\mu = 1.$$

Write $t_j(x, y) = |\langle x_j, y_j \rangle|$. The quadratic tensor energy satisfies

$$Q(\mu) := \iint \prod_{j=1}^{m} t_j(x, y)^2 \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) = \operatorname{tr}(T_\mu^2) \geq \frac{1}{D}, \tag{4.1}$$

with equality if and only if $T_\mu = I_H/D$. The inequality follows from Cauchy–Schwarz on the D eigenvalues of T_μ.

Proposition 4.1 (Strict quadratic certificate)**.** *Let $h_j : [0, 1] \to [0, \infty)$ be continuous functions, with $h_j(0) = 0$, $c_j = h_j(1) > 0$, and*

$$h_j(t) \geq c_j t^2 \quad (0 \leq t \leq 1), \qquad h_j(t) > c_j t^2 \quad (0 < t < 1). \tag{4.2}$$

Let $K(x, y) = \prod_j h_j(t_j(x, y))$. Then

$$\min_{\mu \in \mathcal{P}(X)} \iint K \, \mathrm{d}\mu\mathrm{d}\mu = \frac{\prod_j c_j}{D}. \tag{4.3}$$

A probability measure minimizes this energy if and only if its projective tensor support consists of exactly D pairwise orthogonal lines spanned by product tensors, each with mass $1/D$. In particular every minimizer has at most $2^m D$ atoms.

Proof. Let $C = \prod_j c_j$. By the product of the inequalities in (4.2) and (4.1), the energy is at least $CQ(\mu) \geq C/D$. To attain the bound, choose an orthonormal basis $e_{j,1}, \ldots, e_{j,d_j}$ of each $\mathbb{R}^{d_j}$ and distribute equal mass over the D Cartesian products of basis vectors. Each self-interaction contributes C, and each interaction between distinct products vanishes, so the energy is C/D.

If μ minimizes, then $T_\mu = I_H/D$ and

$$\iint \Big(K(x, y) - C \prod_j t_j(x, y)^2 \Big) \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) = 0.$$

The integrand is continuous and nonnegative. By Lemma 2.4 it vanishes for every pair x, y in $S = \operatorname{supp} \mu$. If some $t_j = 0$, then $z(x) \perp z(y)$. Otherwise all $t_j > 0$, and strictness in (4.2) implies that all $t_j = 1$. In this latter case the projective lines $\mathbb{R}z(x)$ and $\mathbb{R}z(y)$ coincide. Thus distinct projective tensor lines occurring in S are pairwise orthogonal, so there are at most D of them. Since T_μ has rank D, there must be exactly D, and its equality to I_H/D forces mass $1/D$ on each line.

Lemma 2.6 shows that each projective tensor line has at most 2^m representatives in X, so S is finite with the claimed bound. Conversely, a measure with the stated projective-line structure and masses has $T_\mu = I_H/D$ and only zero or unit overlaps, giving energy C/D. □

Remark 4.2. The D orthogonal product tensors in Proposition 4.1 need not form a Cartesian product of individual orthonormal bases. The appropriate classification is an *orthonormal product-tensor basis*, allowing its finitely many projective sign lifts.

5. A sharp mean-exponent criterion, including powers above 2

For σ as in (1.4), write $c = c_\sigma$ and $\bar{p} = \bar{p}_\sigma$. The following elementary lemma is the key calculation.

Lemma 5.1 (Sharp power-mixture quadratic criterion)**.** *Suppose σ is a finite nonzero positive measure on $(0, \infty)$ with finite first moment. Then*

$$h_\sigma(t) \geq ct^2 \qquad (0 \leq t \leq 1) \tag{5.1}$$

holds if and only if $\bar{p} \leq 2$. If $\bar{p} \leq 2$, the inequality is strict for all $0 < t < 1$ unless $\sigma = c\delta_2$.

Proof. For $0 < t < 1$, divide by ct^2 and apply Lemma 2.5 to the random variable $Z = (p-2)\log t$ under the probability measure σ/c:

$$\frac{h_\sigma(t)}{ct^2} = \frac{1}{c}\int t^{p-2}\,\mathrm{d}\sigma(p) \geq t^{\bar{p}-2} \geq 1 \quad (\bar{p} \leq 2). \tag{5.2}$$

The first inequality is strict unless the exponent is constant σ-almost surely; the second is strict if $\bar{p} < 2$. When $\bar{p} = 2$, equality for an interior t therefore forces $\sigma = c\delta_2$. At $t = 0, 1$ equality holds by the definitions.

For necessity, the left derivative of h_σ at 1 is $\int p\,\mathrm{d}\sigma(p)$, by dominated convergence and the finite first moment. If $h_\sigma(t) - ct^2 \geq 0$ on $[0,1]$ and equals 0 at 1, its derivative at 1 from the left must be nonpositive. Hence $c\bar{p} - 2c \leq 0$, that is $\bar{p} \leq 2$. □

Proof of Theorem 1.2. Lemma 5.1 shows that each factor h_{σ_j} satisfies the non-strict quadratic inequality. Thus (4.1) gives the lower bound $\prod c_j/D$, attained by orthonormal product-basis measures. If no σ_j is concentrated at $p = 2$, Lemma 5.1 gives strictness in every coordinate, so Proposition 4.1 yields the complete finite-support classification.

Conversely, suppose $\sigma_k = c_k\delta_2$. Choose the normalized rotation-invariant probability on $\mathbb{S}^{d_k-1}$ in coordinate k (explicitly constructed in the proof of Proposition 7.1), and uniform distributions on orthonormal bases in all other coordinates, independently. Rotational invariance of a distribution ω_d forces its matrix $\int xx^{\mathsf{T}}\,\mathrm{d}\omega_d(x)$ to have vanishing off-diagonal entries (by reflections) and equal diagonal entries (by coordinate permutations). Since its trace equals one, it is I_d/d. This measure has infinite support because $d_k \geq 2$. Its tensor second moment is I_H/D. For a pair in its support, either a discrete coordinate differs and both K and $C\prod_j t_j^2$ vanish, or all discrete coordinates agree, and the only nontrivial factor is $h_{\sigma_k}(t) = c_k t^2$. Hence its energy is exactly C/D, making it an infinitely supported minimizer. □

Corollary 5.2 (Two-power superquadratic interactions)**.** *Let $0 < q_j < 2 < r_j$ and let $a_j, b_j > 0$ satisfy*

$$b_j(r_j - 2) \leq a_j(2 - q_j) \qquad (j = 1, \ldots, m). \tag{5.3}$$

For

$$K(x,y) = \prod_{j=1}^{m} \left(a_j t_j(x,y)^{q_j} + b_j t_j(x,y)^{r_j}\right),$$

we have

$$\min_\mu \iint K\,\mathrm{d}\mu\mathrm{d}\mu = \frac{\prod_j (a_j + b_j)}{D},$$

and every minimizer has the finite-support classification of Proposition 4.1. The threshold (5.3) *is necessary and sufficient for the displayed* coordinatewise quadratic minorant.

Proof. Use the exponent measure $\sigma_j = a_j\delta_{q_j} + b_j\delta_{r_j}$. Its mean is at most 2 precisely under (5.3); it is not concentrated at 2. The conclusion follows from Theorem 1.2 and Lemma 5.1. □

Example 5.3 (Exact rational superquadratic value)**.** Set $(d_1, d_2) = (2, 3)$ and

$$K(x,y) = \left(t_1 + \frac{2}{3}t_1^{7/2}\right)\left(t_2^{3/2} + \frac{3}{10}t_2^{11/3}\right), \qquad t_j = |\langle x_j, y_j\rangle|.$$

The weighted mean exponent in each factor is exactly 2:

$$\frac{1 + (2/3)(7/2)}{1 + 2/3} = 2, \qquad \frac{3/2 + (3/10)(11/3)}{1 + 3/10} = 2.$$

The criterion is satisfied *at equality*, and all minimizers are finite. With $D = 6$,

$$\boxed{\min_\mu \iint K(x,y)\,\mathrm{d}\mu(x)\mathrm{d}\mu(y) = \frac{(5/3)(13/10)}{6} = \frac{13}{36}.} \tag{5.4}$$

Every minimizer occupies exactly six orthogonal projective product-tensor lines, with mass $1/6$ per line and at most 24 support points on $\mathbb{S}^1 \times \mathbb{S}^2$. The standard product-basis measure provides an explicit minimizer.

6. Correlated exponent mixtures: a cone criterion

Factorization of the exponent law is not required to obtain exact values. Let ρ be a finite nonzero positive measure on $(0,\infty)^m$, with total mass A and finite first moments, and set

$$K_\rho(x,y) = \int \prod_{j=1}^{m} t_j(x,y)^{p_j}\,\mathrm{d}\rho(\mathbf{p}), \quad q_j = A^{-1}\int p_j\,\mathrm{d}\rho(\mathbf{p}). \tag{6.1}$$

Theorem 6.1 (Joint-law quadratic certificate and strict contact)**.** *Assume $q_j \le 2$ for each j. Then*

$$\min_{\mu\in\mathcal{P}(X)} \iint K_\rho\,\mathrm{d}\mu\mathrm{d}\mu = \frac{A}{D}. \tag{6.2}$$

Suppose additionally that for every nonzero $s \in [0,\infty)^m$, at least one of the following holds:

(i) $\sum_j s_j(2-q_j) > 0$;

(ii) *the random variable $\sum_j s_j p_j$ is not constant ρ-almost everywhere.*

Then every minimizer is finitely supported and has exactly the projective product-tensor classification in Proposition 4.1. The additional condition is equivalent to strictness of the quadratic pointwise certificate whenever every $t_j > 0$ and at least one $t_j < 1$.

Proof. If all $t_j > 0$, set $s_j = -\log t_j \ge 0$. Lemma 2.5 gives

$$\begin{aligned}\frac{K_\rho(x,y)}{A\prod_j t_j^2} &= \frac{1}{A}\int \exp\left(-\sum_j s_j(p_j-2)\right)\mathrm{d}\rho(\mathbf{p}) \\ &\ge \exp\left(\sum_j s_j(2-q_j)\right) \ge 1.\end{aligned} \tag{6.3}$$

If some $t_j = 0$, both sides of $K_\rho \ge A\prod_j t_j^2$ vanish. Applying (4.1) proves the lower bound A/D, attained by a uniform orthonormal product-basis measure.

For all positive t_j, equality in (6.3) holds exactly when both (i) $\sum_j s_j(2-q_j) = 0$ and (ii) $\sum_j s_j p_j$ is ρ-almost surely constant. The additional hypothesis excludes these simultaneous conditions whenever $s \ne 0$. Thus the inequality is strict unless all $t_j = 1$. The argument of Proposition 4.1 applies verbatim, giving the classification. The same equality conditions establish the final equivalence. □

Proposition 6.2 (Necessity of the mean-vector condition for the joint pointwise certificate)**.** *Assume ρ has finite first moments. The inequality*

$$\int \prod_j t_j^{p_j}\,\mathrm{d}\rho(\mathbf{p}) \ge A\prod_j t_j^2 \quad \textit{for all } (t_1,\dots,t_m)\in[0,1]^m$$

holds if and only if $q_j \le 2$ for every j.

Proof. The sufficient direction is inequality (6.3), including the case of a zero coordinate. For necessity, fix k, put $t_j = 1$ when $j \ne k$, and let $t_k = t$. We obtain $\int t^{p_k}\,\mathrm{d}\rho \ge At^2$. Both sides agree at $t = 1$. Their left derivatives there exist by the finite first-moment assumption, so the derivative of the left-hand difference must be nonpositive: $Aq_k - 2A \le 0$. □

Example 6.3 (A correlated superquadratic law with complete discreteness)**.** Take $m = 2$, $(d_1,d_2) = (2,3)$, and $\rho = \frac{1}{2}\delta_{(1,3)} + \frac{1}{2}\delta_{(2,1)}$. Then $A = 1$ and $(q_1,q_2) = (3/2,2)$. For every nonzero $s = (s_1,s_2) \ge 0$, either $s_1 > 0$, so $(2-q)\cdot s = s_1/2 > 0$, or $s_1 = 0 < s_2$, in which case $s\cdot\mathbf{p}$ takes the distinct values $3s_2$ and s_2. Hence the strict-contact hypothesis holds, and

$$K(x,y) = \tfrac{1}{2}t_1t_2^3 + \tfrac{1}{2}t_1^2t_2, \qquad \min_\mu I_K(\mu,\mu) = \frac{1}{6}.$$

Every minimizer has exactly six mutually orthogonal projective product-tensor directions, each of mass $1/6$, despite the exponent $3 > 2$ and the perfect dependence of exponent coordinates.

Corollary 6.4 (Subquadratic joint mixtures)**.** *If ρ is supported on $(0,2]^m$ and $\rho\{\mathbf{p} : p_j < 2\} > 0$ for each j, every minimizer is finitely supported with at most $2^m D$ atoms. If this latter coverage condition fails for any j, an infinitely supported minimizer exists. In both cases the minimum is A/D.*

Proof. Under coverage, each $q_j < 2$, so condition (i) in Theorem 6.1 holds for every nonzero $s \geq 0$. Conversely, if coverage fails for coordinate k, then $p_k = 2$ almost everywhere. Use an independent rotation-invariant measure in coordinate k and uniform orthonormal-basis measures in all other coordinates. As in the proof of Theorem 1.2, every mixed kernel agrees with its quadratic lower bound on pairs in the support, and $T_\mu = I_H/D$. This is an infinitely supported minimizer. □

Remark 6.5 (Why the cone hypothesis is not vacuous). For $m = 2$, let $\rho = \delta_{(1,3)} + \delta_{(3,1)}$. Both mean exponents are 2, and the minimum is $A/D = 2/D$. But for any $0 < t < 1$ and $t_1 = t_2 = t$,

$$t_1 t_2^3 + t_1^3 t_2 = 2t^4 = 2t_1^2 t_2^2.$$

Thus strictness fails along the positive diagonal $s_1 = s_2 > 0$. Theorem 6.1 makes no all-minimizer finiteness claim in this example. The failure of this *certificate* should not be confused with proof of an infinitely supported minimizer.

7. The pure rational $p > 2$ boundary and continuous counterexamples

The tensor finite-replacement theorem applies to pure rational exponents, but it does not prove full discreteness. For the single sphere, Theorem 1.1 yields

$$\#\operatorname{supp}\nu \leq \begin{cases} \binom{d+a-1}{a}, & a \text{ even}, \\ \binom{d+2a-1}{2a}, & a \text{ odd}, \end{cases} \quad \text{when } p = a/b > 0 \text{ is reduced.} \tag{7.1}$$

Park [8] reports the stronger implication $b \geq d \Rightarrow$ every minimizer finite, based on additional tools not reconstructed here. The cases $b < d$ for general non-even $p > 2$ are not resolved by our arguments.

One cannot claim that *all* positive rational $p > 2$ produce finite original minimizers: positive even exponents provide counterexamples.

Proposition 7.1 (Infinitely supported minimizers for pure even powers)**.** *Suppose $p_j = 2k_j$ with integers $k_j \geq 1$. Then the product of rotation-invariant probability measures on the spheres is a minimizing measure for $\mathcal{E}_\mathbf{p}$, and it has infinite support.*

Proof. The kernel has a finite-dimensional positive semidefinite feature representation

$$K_\mathbf{p}(x,y) = \langle \Psi(x), \Psi(y) \rangle, \qquad \Psi(x) = \bigotimes_j x_j^{\otimes 2k_j}.$$

Therefore $\mathcal{E}_\mathbf{p}(\mu) = \| \int \Psi \, \mathrm{d}\mu \|^2$. For completeness we construct the required rotation average without assuming a Haar-measure theorem. For each $d \geq 2$, let B be a $d \times d$ random matrix with independent standard normal entries and put $U = B(B^\mathsf{T} B)^{-1/2}$; almost surely B is invertible and U is orthogonal. Since the Gaussian density depends only on $\operatorname{tr}(B^\mathsf{T} B)$, the laws of QB and BQ coincide with that of B for each fixed orthogonal Q. The polar-factor formula shows that the law of U is invariant under both left and right orthogonal multiplication. For any fixed unit x, Ux consequently has a distribution independent of x and invariant under all rotations; call it ω_d. This measure has full spherical support and no atoms (its orbit under rotations is infinite, and invariant point masses would otherwise have infinite total mass).

Take independent such random orthogonal matrices U_j for all factors. For any μ on X, push μ forward by the independent coordinatewise rotations and average; by right invariance of each U_j the result is $\omega = \bigotimes_j \omega_{d_j}$, independent of the starting μ. On the feature space, the resulting action W_U is orthogonal. If $v = \int \Psi \, \mathrm{d}\mu$, the elementary variance identity gives

$$\mathcal{E}_\mathbf{p}(\omega) = \|\mathbb{E}_U W_U v\|^2 \leq \mathbb{E}_U \|W_U v\|^2 = \|v\|^2 = \mathcal{E}_\mathbf{p}(\mu).$$

Thus ω minimizes, and it is infinitely supported because $d_j \geq 2$. □

Example 7.2 (An exact continuous superquadratic minimizer)**.** On the circle $\mathbb{S}^1 \subset \mathbb{R}^2$ with $p = 4$, rotation invariance gives

$$\mathcal{E}_4(\omega) = \frac{1}{2\pi} \int_0^{2\pi} \cos^4 \theta \, \mathrm{d}\theta = \frac{3}{8}.$$

Proposition 7.1 certifies that this is the minimum, although ω is not finitely supported. Equal masses on the three projective lines at angles 0, $\pi/3$ and $2\pi/3$ also attain $3/8$, since the six off-diagonal ordered interactions equal $(1/2)^4 = 1/16$. This illustrates concretely why "a finite minimizer exists" differs from "every minimizer is finite."

Remark 7.3 (Mathematical scope and external literature). Our complete-discreteness results in Sections 5–6 rely on a lower-order component or on a joint-law quadratic certificate, and therefore do *not* resolve the pure non-even superquadratic problem. The paper [8] is cited for its stated abstract-level conclusions. Determining whether the particular tensor or exponent-law formulations here are genuinely new requires a systematic independent literature and priority review. The mathematics presented here is intended to be auditable, not represented as peer-reviewed or arXiv-accepted.

8. Independent equality checks and computational audit

Every equality used above can be checked by finite-dimensional calculations. We detail the checks to make the examples reproducible.

8.1. **The tensor trace identity.** With $z = z(x)$, a rank-one matrix has entries $(zz^{\mathsf{T}})_{ab} = z_a z_b$ in an orthonormal basis. Thus

$$\begin{aligned}\operatorname{tr}(T_\mu^2) &= \operatorname{tr}\left(\iint z(x)z(x)^{\mathsf{T}} z(y)z(y)^{\mathsf{T}} \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) \right) \\ &= \iint (z(x)^{\mathsf{T}} z(y))^2 \, \mathrm{d}\mu(x)\mathrm{d}\mu(y) = \iint \prod_{j=1}^{m} t_j(x,y)^2 \, \mathrm{d}\mu(x)\mathrm{d}\mu(y).\end{aligned}$$

Since the nonnegative eigenvalues $\lambda_1, \ldots, \lambda_D$ of T_μ add to one,

$$0 \leq \sum_{i=1}^{D} (\lambda_i - 1/D)^2 = \operatorname{tr}(T_\mu^2) - 1/D.$$

Equality holds exactly when every eigenvalue equals $1/D$, which is equivalent to $T_\mu = I_H/D$.

8.2. **The uniform product-basis candidate.** For any choice of orthonormal bases $(e_{j,k})_{k=1}^{d_j}$, define

$$\mu_0 = \frac{1}{D} \sum_{k_1=1}^{d_1} \cdots \sum_{k_m=1}^{d_m} \delta_{(e_{1,k_1}, \ldots, e_{m,k_m})}.$$

Two distinct multi-indices differ in at least one coordinate, yielding $K(x,y) = 0$ for every kernel used in the exact-minimum theorems. A point compared with itself has $K(x,x) = C$, the diagonal constant ($C = \prod_j c_j$ in the factorized case and $C = A$ in the joint case). Exactly D of the D^2 ordered pairs are diagonal, all with mass $1/D^2$. Therefore

$$I_K(\mu_0, \mu_0) = D \frac{C}{D^2} = \frac{C}{D}, \qquad T_{\mu_0} = \frac{1}{D} I_H.$$

This verifies the upper bounds and equality without approximation.

8.3. **Auditing the rational examples.** For the $R = 143$ example, the reduced fractions $5/2$ and $3/1$ require L divisible by 4 and 2, respectively. Thus $L = 4$, $n_1 = 10$, $n_2 = 12$, and

$$R = \binom{2+10-1}{10}\binom{2+12-1}{12} = 11 \cdot 13 = 143.$$

For the 13/36 example, the factor exponent means satisfy

$$\frac{1+(2/3)(7/2)}{5/3} = \frac{1+7/3}{5/3} = 2, \qquad \frac{3/2+(3/10)(11/3)}{13/10} = \frac{3/2+11/10}{13/10} = 2.$$

Both measures have two distinct support exponents and hence strict quadratic contact for all interior overlaps. The diagonal constants are exactly $5/3$ and $13/10$, and hence

$$I_K(\mu_0, \mu_0) = \frac{(5/3)(13/10)}{2 \cdot 3} = \frac{13}{36}.$$

8.4. **The continuous $p = 4$ counterexample.** For $x = (\cos\theta, \sin\theta)$, uniform θ on $[0, 2\pi)$ gives for independent x, y

$$I_4(\omega) = \frac{1}{2\pi}\int_0^{2\pi} \cos^4\theta \, \mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} \frac{3 + 4\cos 2\theta + \cos 4\theta}{8} \, \mathrm{d}\theta = \frac{3}{8}.$$

For three projective lines at angular coordinates $0, \pi/3, 2\pi/3$ with equal weights, the three diagonal ordered pairs have interaction 1, while the six off-diagonal ordered pairs have interaction $(1/2)^4 = 1/16$. Hence

$$I_4 = \frac{3}{9} + \frac{6}{9 \cdot 16} = \frac{1}{3} + \frac{1}{24} = \frac{3}{8}.$$

By the independently proved rotational-averaging argument, $3/8$ is the global minimum, not merely a numerical coincidence.

9. Why there is no Nash curve selection in our arguments

9.1. **A methodological distinction, not a prohibition.** The absence of Nash curve selection in Theorems 1.1, 1.2, and 6.1 is intentional and mathematically consequential. It is *not* a statement that Nash curve selection is unavailable on a product of spheres, nor that a multivariable problem cannot have algebraic arcs. The actual reason is that these theorems prove their conclusions by two elementary mechanisms which do not require converting a possibly infinite support into a curve:

(1) The rational finite-replacement argument establishes a uniform *rank bound*, after which a finite-dimensional convex-geometric barycenter theorem produces one finite optimizer. It says nothing by itself about the number of points originally in the support.
(2) The quadratic-certificate argument proves a *nonnegative pointwise gap* with precisely described zero set. An energy minimizer must have every pair of support points in that zero set. The latter forces mutual orthogonality of distinct projective tensor directions, so finiteness follows from linear algebra without any algebraic selection procedure.

The first mechanism works for *pure rational exponents above two* but yields only finite replacement. The second mechanism yields finiteness of *every* minimizer but requires an additional coordinatewise or joint quadratic certificate, which generally fails for a pure exponent $p > 2$.

Here the term *Nash curve selection* means a result from real algebraic geometry: at a nonisolated point of a semialgebraic set, one can select a nonconstant real-analytic semialgebraic arc approaching that point. A Nash arc is analytic and its coordinate germs are algebraic over the parameter field. This is unrelated to a Nash equilibrium in game theory, and it must not be confused with Carathéodory's finite convex-combination theorem.

9.2. **Exactly where a curve is needed in the single-sphere theorem.** The 2026 single-sphere theorem of Park [8] has a conclusion strictly stronger than finite replacement. Writing $p = a/b$ in lowest terms, it proves that every minimizing measure on $\mathbb{S}^{d-1}$ is finitely supported when $b \geq d$. We do not claim to prove that external theorem here; instead we spell out the geometric obstacle it overcomes, following its stated construction.

Suppose an infinite compact $S \subset \mathbb{S}^{d-1}$ carries a positive semidefinite restriction of the kernel $K_p(x, y) = |\langle x, y\rangle|^{a/b}$. A finite Gram representation gives $h : S \to \mathbb{R}^r$ with

$$\langle h(x), h(y)\rangle = |\langle x, y\rangle|^{a/b}, \qquad \|h(x)\| = 1.$$

In the product ambient space, consider the feature graph

$$A = \{(x, h(x)) : x \in S\} \subset \mathbb{R}^d \times \mathbb{R}^r.$$

Let Z denote its real Zariski closure, the simultaneous real zero set of all real polynomials that vanish on A. The crucial polynomial identity is

$$(u \cdot v)^{2b} = (x \cdot y)^{2a}, \qquad (x, u), (y, v) \in Z. \tag{9.1}$$

It holds first on $A \times A$, and extends to $Z \times Z$ by fixing one argument at a time and using polynomial vanishing. Moreover, Z lies on the two unit spheres and hence is compact. The fibers of projection $(x, u) \mapsto x$ have at most two points because (9.1) forces $u \cdot v = \pm 1$ when the two x-coordinates coincide.

An infinite compact real algebraic set has a nonisolated point; Nash curve selection on its complement at that point yields a nonconstant Nash arc $t \mapsto (x(t), u(t))$ in Z. Since projection has finite fibers, the x-coordinate is nonconstant. The Nash property allows passage to the normalization of the complex algebraic curve traced by the arc. On the resulting compact Riemann surface, the coordinate functions are meromorphic. If

$$A_t(s) = x(s) \cdot x(t), \qquad B_t(s) = u(s) \cdot u(t),$$

then (9.1) gives

$$B_t(s)^{2b} = A_t(s)^{2a}.$$

Taking vanishing orders at any point of the curve yields

$$b \operatorname{ord} B_t = a \operatorname{ord} A_t, \qquad b \mid \operatorname{ord} A_t, \tag{9.2}$$

where the second conclusion uses $\gcd(a, b) = 1$. The coefficient functions A_t lie in the span of at most d meromorphic coordinate functions. The *Wronskian multiplicity bound* for such a finite-dimensional meromorphic function space says that outside a fixed finite exceptional set a nonzero member can have zero order at most $d-1$. When $b \geq d$, the divisibility (9.2) excludes all such zeros. A divisor argument then forces the projective family $[A_t]$ to be constant, contradicting the nonconstant projected curve. For the full divisor-theoretic proof, see [8, Section 3].

Two logical caveats are essential. First, an arbitrary infinite compact S need not contain any arc at all: a Cantor-type compact set is an elementary counterexample. The arc is selected in the *algebraic feature lift* Z, not asserted to lie in S. Second, the algebraic curve exists for the purpose of a *contradiction to infinite support*. A rank bound alone would not produce that contradiction.

9.3. **What the multivariable polynomial lift really gives.** For the rational tensor-product kernel from Theorem 1.1, let L and the even integers n_j be as in (1.3). On a finite-dimensional feature lift, the polynomial relation is

$$(u \cdot v)^L = \prod_{j=1}^{m} (x_j \cdot y_j)^{n_j}. \tag{9.3}$$

The same real-algebraic lifting idea is, in principle, available. But after selecting and normalizing an algebraic arc, the order identity becomes

$$L \operatorname{ord} B_t = \sum_{j=1}^{m} n_j \operatorname{ord} A_{j,t}, \quad A_{j,t}(s) = x_j(s) \cdot x_j(t). \tag{9.4}$$

This is a *sum of orders*, whereas the single-sphere identity (9.2) is a divisibility constraint on *one* meromorphic function. In general,(9.4) permits simultaneous zeros in different factors to offset each other's arithmetic obstructions. It does not imply $b_j \mid \operatorname{ord} A_{j,t}$ for each j.

For an explicit arithmetic audit take

$$(p_1, p_2) = (1/2, 1/2), \quad L = 4, \quad (n_1, n_2) = (2, 2).$$

The order identity is

$$4 \operatorname{ord} B_t = 2 \operatorname{ord} A_{1,t} + 2 \operatorname{ord} A_{2,t},$$

so only the *sum* of the two right-hand orders is forced to be even. Two simple zeros of order one are compatible with this identity; there is no factorwise even-order conclusion. This is an illustration of the algebraic obstruction only, *not* an example of an infinitely supported optimal measure.

A second difference is that a selected curve can vary only in a subset of the factors. If $x_k(t)$ is constant on the curve, then $A_{k,t}(s)$ is a nonzero constant for regular s, t, and the curve contains no multiplicity information about the denominator b_k. Finally, the spaces of functions $A_{j,t}$ have different dimensions (up to d_j); the combined product identity does not automatically produce a single d-dimensional linear series to which the single-sphere Wronskian argument can be applied.

Thus the fact that Nash selection is *absent here* reflects a choice of proof technique, while the lack of an immediate coordinatewise denominator theorem reflects a *genuine gap in transferring the subsequent order-of-vanishing argument*. We neither assert that such a multivariable theorem is false nor that a more sophisticated algebraic approach cannot prove it.

9.4. **What replaces curve selection in the strong classification.** The mixed-energy conclusions use a different route. Let $t_j = |\langle x_j, y_j \rangle|$, $C = \prod c_j$, and assume the strict coordinatewise certificate

$$h_j(t) \geq c_j t^2, \qquad h_j(t) > c_j t^2 \quad \text{for } 0 < t < 1.$$

The matrix T_μ in Section 4 satisfies

$$I_K(\mu) \geq C \operatorname{tr}(T_\mu^2) \geq C/D,$$

with the value C/D attained by a product orthonormal-basis measure. For every minimizing μ, both inequalities are equalities. Since the continuous contact gap

$$G(x, y) = K(x, y) - C \prod_j t_j(x, y)^2$$

is nonnegative and has zero $\mu \otimes \mu$ integral, it vanishes *on the entire support square*, not merely almost everywhere. Therefore two support points either have $t_j = 0$ for some factor, or have $t_j = 1$ for every factor. The corresponding rank-one tensors are respectively orthogonal or represent the same projective line. As $\dim H = D$, there are at most D such lines. The full-rank equality $T_\mu = I_H/D$ forces exactly D lines of mass $1/D$. No attempt to select an arc or bound analytic multiplicities is necessary: infinite support is excluded directly by an exact equality set and linear algebra. The same reasoning works for $m = 1$ whenever a strict quadratic certificate exists, so it would be incorrect to say that *multivariable structure itself* makes Nash selection unnecessary.

10. Single-sphere versus product-sphere versus genuine k-point energies

10.1. **Three definitions which must not be conflated.** The word "multivariable" has two different conventional meanings in this literature. In order to compare results correctly, distinguish:

(1) *Single-sphere, pairwise p-frame energy:* a single $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and

$$I_p(\mu) = \iint |\langle x, y\rangle|^p \, \mathrm{d}\mu(x)\mathrm{d}\mu(y).$$

There are two independently sampled particles and one inner product.

(2) *Product-sphere, pairwise tensor energy (this paper):* $\mu \in \mathcal{P}(\prod_j \mathbb{S}^{d_j-1})$ and

$$I_{\mathbf{p}}(\mu) = \iint \prod_j |\langle x_j, y_j\rangle|^{p_j} \, \mathrm{d}\mu(x)\mathrm{d}\mu(y).$$

Again there are exactly *two* independently sampled particles, but each particle has m coordinates. The coordinates of a particle can be correlated under μ. The exponent vector may also be random or correlated in the mixture generalizations.

(3) *Genuine k-point multivariate energy, $k \geq 3$:* for a symmetric kernel $\mathcal{K} : X^k \to \mathbb{R}$,

$$\mathfrak{I}_{\mathcal{K}}(\mu) = \int_{X^k} \mathcal{K}(x^{(1)}, \ldots, x^{(k)}) \, \mathrm{d}\mu(x^{(1)}) \cdots \mathrm{d}\mu(x^{(k)}).$$

This is the setting called multivariate kernel potential theory in [4, 5]. Its interactions involve k distinct sampling slots, and the energy is a degree-k functional of μ rather than a quadratic functional.

Our theorems concern item 2; they are *not* automatically extensions of the k-point theorems from item 3.

10.2. **A comparison of rigorously established conclusions.**

Question	Single sphere, pairwise I_p	Pairwise tensor product, this paper
Ambient space	$\mathbb{S}^{d-1}$	$X = \prod_j \mathbb{S}^{d_j-1}$; $D = \prod_j d_j$
Even exponent obstruction	For $p = 2k$, a rotation-invariant continuous minimizer exists.	For all $p_j \in 2\mathbb{N}$, a product rotation-invariant continuous minimizer exists.
Subquadratic regime	If $0 < p < 2$, all minimizers lie on d orthogonal projective lines, each of mass $1/d$.	For strict quadratic certificates, all minimizers lie on D orthogonal projective tensor lines, each of mass $1/D$.
Rational exponents	A finite replacement exists; Park proves all-minimizer finiteness if reduced $b \geq d$.	A finite replacement exists with bound R; no coordinatewise $b_j \geq d_j$ implication is proved here.
Pure $p > 2$	The all-minimizer question remains open in general for non-even rational p with $b < d$ and for arbitrary irrational p.	The pure tensor all-minimizer problem is likewise not settled by the rational replacement theorem.
Main strong method	For rational $b \geq d$: Nash arc and Wronskian rigidity in Park's theorem.	For power mixtures: quadratic minorant, contact zero set, and dimension of H.
Support cardinality	Projective mass on d lines for $0 < p < 2$; sphere support at most $2d$.	Projective mass on D lines under strict certificate; sphere-product support at most $2^m D$.

Here a finite *replacement* always means a new optimal measure supported within the support of the original; it does not assert that the original measure is discrete.

10.3. **Product measures, correlated measures, and a non-Cartesian example.** For a pure tensor kernel, if $\mu = \mu_1 \otimes \cdots \otimes \mu_m$, Fubini's theorem gives the exact factorization

$$I_{\mathbf{p}}(\mu_1 \otimes \cdots \otimes \mu_m) = \prod_{j=1}^{m} I_{p_j}(\mu_j). \tag{10.1}$$

Writing $e_{d_j}(p_j)$ for the individual single-sphere minima, we consequently have

$$\min_{\mu\in\mathcal{P}(X)} I_{\mathbf{p}}(\mu) \le \prod_{j=1}^{m} e_{d_j}(p_j). \tag{10.2}$$

The inequality points in this direction because the left-hand problem allows *all correlated probability measures*, of which product measures form only a subclass. The product formula by itself does not prove equality for arbitrary exponent vectors, nor does it classify all optimizers.

Even when the exact quadratic theorem applies, a minimizing measure need not be a Cartesian product of independent coordinate laws. For example take $m = 2$, $d_1 = d_2 = 2$. Let e_1, e_2 be the standard basis of $\mathbb{R}^2$ and

$$v_+ = (e_1 + e_2)/\sqrt{2}, \qquad v_- = (e_1 - e_2)/\sqrt{2}.$$

The four product tensors

$$e_1 \otimes e_1, \quad e_1 \otimes e_2, \quad e_2 \otimes v_+, \quad e_2 \otimes v_- \tag{10.3}$$

form an orthonormal basis of $\mathbb{R}^2 \otimes \mathbb{R}^2$, but not one Cartesian product of fixed orthonormal bases in the two factors. The probability measure assigning mass $1/4$ to the corresponding four pairs in $\mathbb{S}^1 \times \mathbb{S}^1$ minimizes every strict-certificate kernel of Proposition 4.1 because distinct basis tensors have a zero coordinate overlap, and identical ones have unit overlaps. The conditional law of the second coordinate depends on the first, so this minimizing measure is correlated. This demonstrates why "orthogonal projective product-tensor basis" is the correct statement and "uniform Cartesian product basis" would be an incorrect classification of *all* minimizers.

10.4. **Why the k-point version is a genuinely different problem.** Consider a genuine symmetric k-point kernel $\mathcal{K}$, with $k \ge 3$, and a finite signed measure η of total mass zero. The expansion of the energy about μ is

$$\mathfrak{I}_{\mathcal{K}}(\mu + t\eta) = \sum_{\ell=0}^{k} \binom{k}{\ell} t^{\ell} \int_{X^k} \mathcal{K}\, \mathrm{d}\eta^{\otimes\ell} \mathrm{d}\mu^{\otimes(k-\ell)}. \tag{10.4}$$

The first variation is a k-fold potential with $k-1$ variables integrated against μ; the second variation is the quadratic form of a *μ-dependent* induced kernel

$$\mathcal{K}_\mu(x, y) = \int_{X^{k-2}} \mathcal{K}(x, y, z_3, \dots, z_k)\, \mathrm{d}\mu(z_3) \cdots \mathrm{d}\mu(z_k).$$

For $k = 2$, the Hessian kernel is the original, fixed interaction kernel, and the energy is exactly quadratic in the measure. This is what permits the support Gram-rank argument in Section 3. For $k \ge 3$, the higher-order terms in (10.4) do not disappear, and a rank bound on $\mathcal{K}_\mu$ would need additional consistency conditions before it could be turned into an optimal replacement. Similarly, the quadratic tensor moment T_μ no longer automatically controls the energy. Therefore neither theorem in this paper should be read as an all-purpose theorem for geometric k-point interactions.

10.5. **The pure $p > 2$ frontier in concrete terms.** Set $m = 1$, $h(t) = t^p$, and $p > 2$. For every $0 < t < 1$,

$$t^p < t^2.$$

Thus the strict quadratic *lower* certificate $h(t) \ge t^2$ used to force finite support is not merely unproved: it is false. The rank/Carathéodory argument still supplies a finite replacement when p is rational, but it has no mechanism to exclude an infinite original support. At even integers such supports actually occur, by Proposition 7.1. For other rational exponents, Park's single-sphere theorem provides a partial structural result when $b \ge d$, with Nash selection being part of the proof; the remaining cases cannot be labeled solved using the present quadratic argument.

The mixed superquadratic theorem is different. For instance, the factor $h(t) = t + (2/3)t^{7/2}$ satisfies the quadratic minorant $h(t) \ge (5/3)t^2$, even though it contains a power above two. Its

proof uses the weighted mean exponent exactly equal to two and a strict convexity argument. The lower-degree term compensates for the higher-degree term. Such compensation is what opens the strict-contact route; it is not a deduction from the pure $p > 2$ problem.

10.6. **A roadmap for a future rational tensor-discreteness theorem.** A result proving finite support of *every* minimizer for pure rational multivariable energies beyond the quadratic-certificate class would require an argument absent from this manuscript. One possible approach would investigate the polynomial feature lift (9.3), stratify Nash arcs by which coordinate inner products vanish, and develop a multiplicity bound for the combined families $A_{j,t}$ that overcomes the coupled order identity (9.4). Another would establish a new direct nondegeneracy or support-separation principle without algebraic curves. These are *research directions*, not theorems or assertions that the proposed strategy will succeed.

The rigorous boundary of the present manuscript is therefore:

> *Finite-rank compression extends to arbitrary rational tensor exponents, but full support discreteness requires an additional mechanism. Our explicit power-mixture certificates supply such a mechanism in a nontrivial superquadratic class. The single-sphere Nash–Wronskian method addresses a different obstruction and cannot be invoked from the tensor rank bound alone.*

11. Precisely delimited conclusions

The results above establish, without reliance on the referenced research literature, all of the following claims:

(1) Each rational-exponent tensor-product energy admits minimizing replacements with at most the explicit number R of atoms, *inside the original minimizing support*; the original measure need not be finite.
(2) Under a coordinatewise mean-exponent condition, every factorized positive power mixture has an exact minimum and, except for pure-quadratic coordinates, a complete classification of *every* minimizer.
(3) Positive exponent-vector laws, without coordinate factorization, admit an exact minimum under a sharp mean-vector quadratic certificate, with a sufficient and checkable strict-contact criterion ensuring finiteness of every minimizer.
(4) Pure positive even exponents permit infinitely supported minimizers; therefore full finite replacement must never be equated with full finiteness.

Sections 9 and 10 explain why these statements use no Nash-curve selection, why the order-of-vanishing part of the single-sphere argument does not transfer automatically, and how pairwise product-sphere kernels differ from genuine k-point interactions.

None of these results proves that every minimizer of the original *pure* kernel $|\langle x, y\rangle|^p$ is finite for every non-even rational $p > 2$. The finite denominator theorem of Park [8] is mentioned as background only. This manuscript neither reproduces that theorem nor makes a priority claim for its tensor-product formulations.

12. Verification summary

The precise reproducible checks are as follows.

(1) **Rational powers:** for $p_j = a_j/b_j$, compute L, n_j, and R using (1.3); the bound follows from Lemmas 3.1 and 3.2, then Lemma 2.3.
(2) **Quadratic optimality:** verify $T_\mu \succeq 0$, $\operatorname{tr} T_\mu = 1$, and $\operatorname{tr} T_\mu^2 \geq 1/D$. Product orthonormal-basis measures attain equality.
(3) **Mean-exponent cutoff:** for each coordinate law, compute $\bar{p} = c^{-1} \int p \, \mathrm{d}\sigma$. Lemma 5.1 gives an if-and-only-if criterion for the pointwise quadratic minorant, with strictness exactly outside the pure quadratic law.

(4) **Every-minimizer finiteness:** check that a continuous nonnegative contact-gap kernel with zero $\mu \otimes \mu$ integral must vanish on $\operatorname{supp}\mu \times \operatorname{supp}\mu$. This forces orthogonal or identical projective tensor lines.

(5) **Exact arithmetic:** in Example 5.3, $D = 2 \cdot 3 = 6$, $c_1 = 5/3$, $c_2 = 13/10$, and $c_1 c_2 / D = 13/36$. The equalities for mean exponents are exact rational identities.

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